ಪಿ.ಡಿ.ಎ. ತಾಂತ್ರಿಕ ಮಹಾವಿದ್ಯಾಲಯ
ಎಐಸಿಟಿ ನವದೆಹಲಿ ಇಂದ ಮಾನ್ಯತೆ ಪಡೆದ – ವ್ಟಿಟಿಯು ಬೆಳಗಾವಿ ಸಂಯೋಜಿತ ಸ್ವಾಯತ್ತ ಸಂಸ್ಥೆ
PDA COLLEGE OF ENGINEERING
Autonomous Institute Affiliated to VTU Belagavi - Approved by AICTE New Delhi

Industrial Internship Report on

"Traffic Forecasting, Mining Quality Prediction & Turbofan RUL Prediction"

Prepared by

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Executive Summary
This report presents the work completed during a 6-week internship starting from 16 February 2026. The internship focused on solving real-world problems using machine learning including traffic forecasting, mining quality prediction, and turbofan engine RUL prediction.

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1 Preface

This report summarizes the work carried out during the 6-week Industrial Internship in the domain of Data Science and Machine Learning, conducted by upskill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT).

In today's competitive environment, internships play a crucial role in bridging the gap between academic knowledge and real-world applications. This internship provided me with an opportunity to gain hands-on experience in solving practical industrial problems using machine learning techniques.

During the internship, I worked on three major projects: Smart City Traffic Forecasting, Mining Quality Prediction, and Turbofan Engine Remaining Useful Life (RUL) Prediction. These projects involved working with real-world datasets and applying concepts such as time series analysis, regression models, and deep learning techniques.

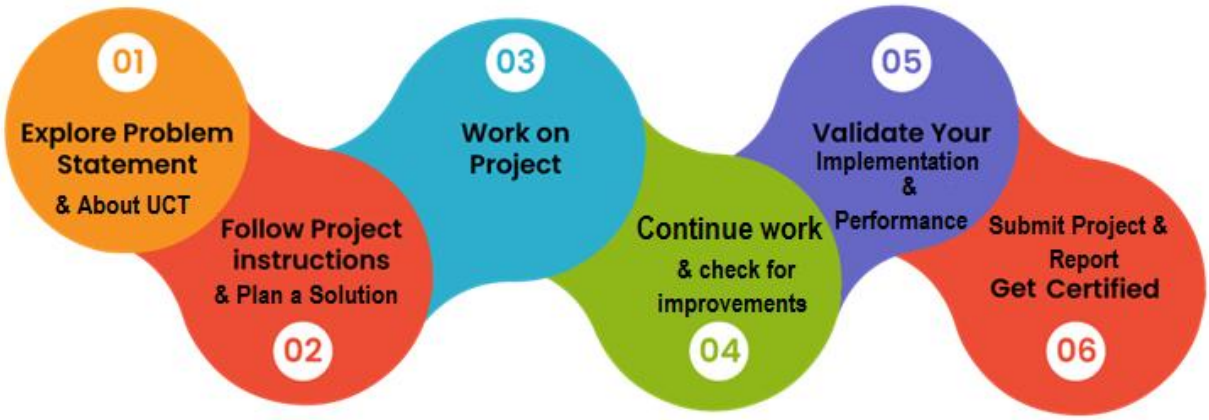
The internship program was well-structured and designed to gradually build understanding from fundamental concepts to real-world implementation. It included learning, experimentation, and project development phases which helped in gaining both theoretical and practical knowledge.

Through this internship, I improved my skills in data preprocessing, feature engineering, model building, and evaluation. I also developed a deeper understanding of how machine learning is applied in industrial scenarios to improve efficiency and decision-making.

I would like to express my sincere gratitude to upskill Campus, UniConverge Technologies Pvt Ltd, and all the mentors who guided and supported me throughout this internship.

Finally, I would like to encourage my juniors and peers to actively participate in such internships, as they provide valuable exposure, enhance technical skills, and prepare us for real-world

challenges.


01
Explore Problem Statement
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Follow Project instructions
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03
Work on Project
04
Continue work
& check for improvements
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Validate Your Implementation
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Performance
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2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

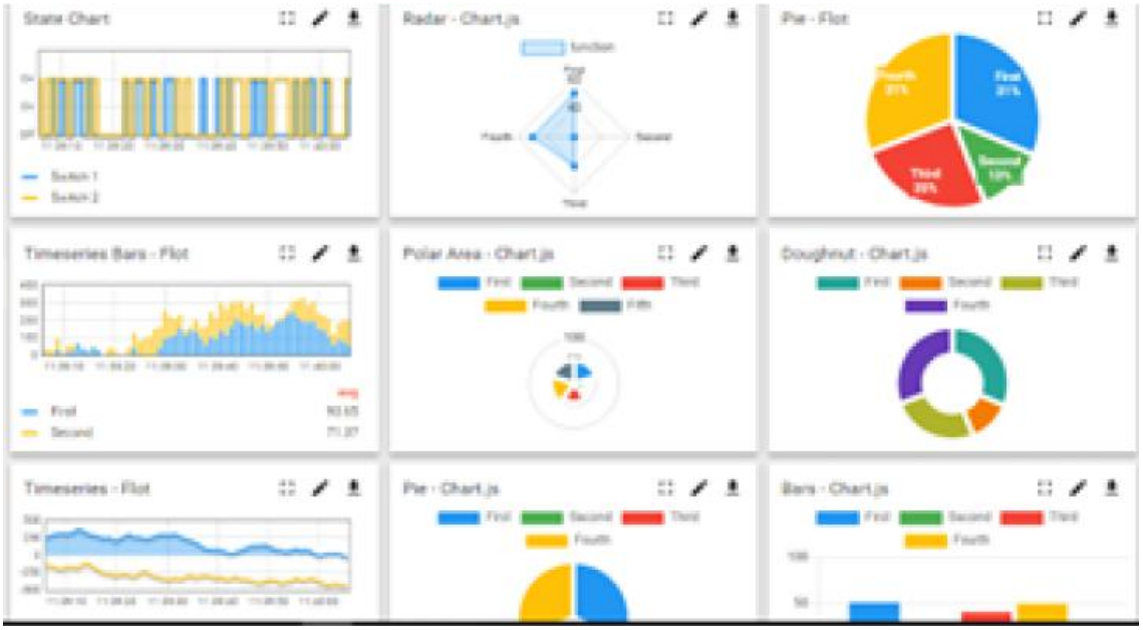
i. UCT IoT Platform (uct Insight)

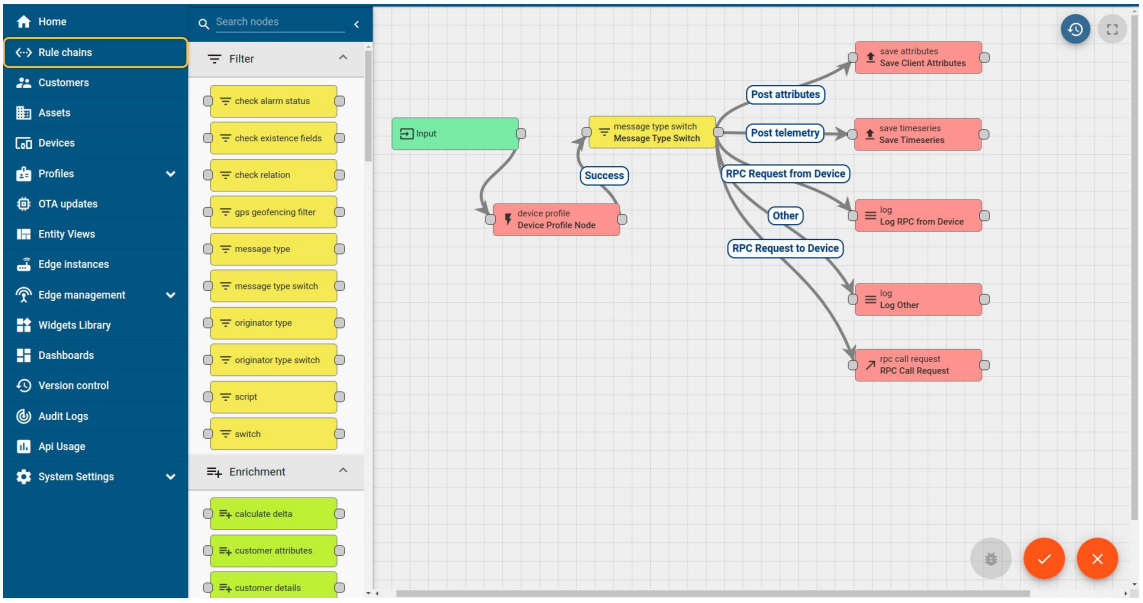
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

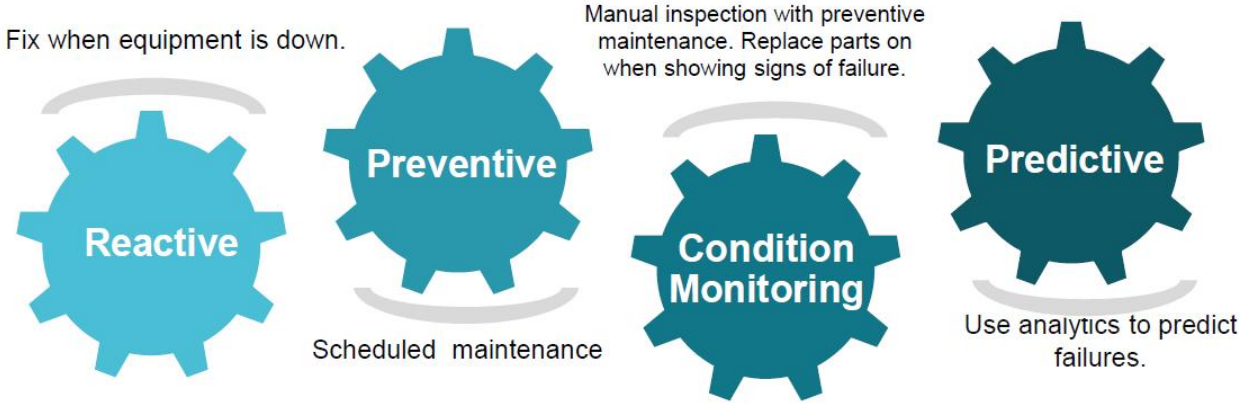
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

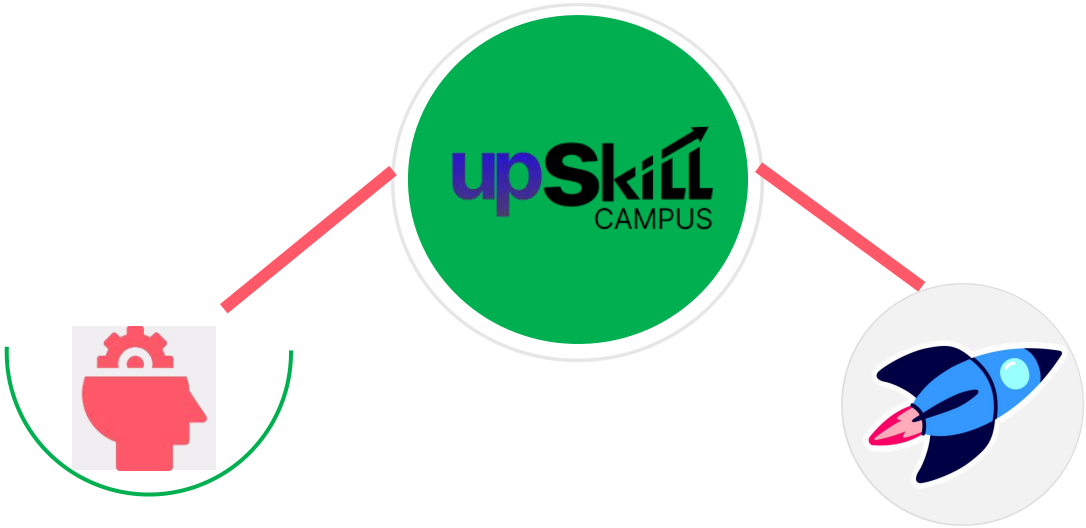
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

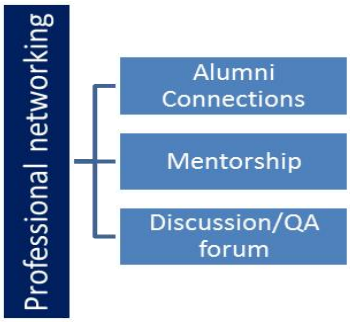
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

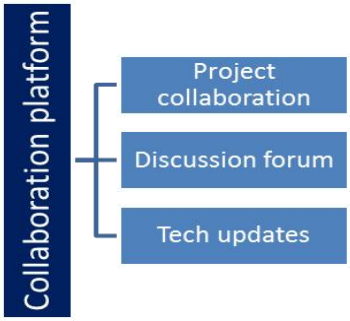


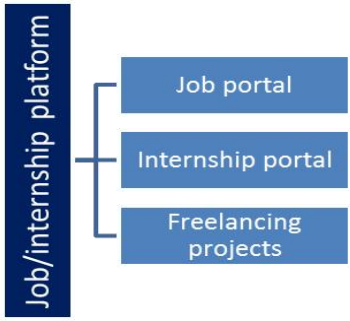
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] NASA C-MAPSS Turbofan Engine Dataset – Prognostics Data Repository

[2] Mining Process Flotation Dataset – Kaggle Dataset

[3] Smart City Traffic Dataset – Provided by Upskill Campus Internship

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Remaining Useful Life	RUL
Long Short-Term Memory	LSTM

Root Mean Squared Error	RMSE
Internet of Things	IOT
Mean Absolute Error	MAE
Deep Learning	DL

3 Problem Statement

In the assigned problem statement, the objective was to solve real-world industrial challenges using Machine Learning techniques.

The first problem involved forecasting traffic patterns in a smart city environment. The goal was to analyze traffic data from multiple junctions and predict future congestion levels, which can help in better traffic management and infrastructure planning.

The second problem focused on predicting the quality of the mining process, specifically the percentage of silica impurity in iron ore concentrate. The aim was to build a predictive model that helps engineers take corrective actions in advance, thereby improving efficiency and reducing environmental impact.

The third problem involved predicting the Remaining Useful Life (RUL) of turbofan engines using sensor data. The objective was to develop a model that can estimate the number of operational cycles before engine failure, enabling predictive maintenance and reducing unexpected breakdowns.

These problems required the application of data preprocessing, feature engineering, and machine learning/deep learning models to generate accurate and reliable predictions.

4 Existing and Proposed solution

Existing Solutions:

Traditional systems for traffic management, mining quality monitoring, and equipment maintenance mainly rely on manual observation and reactive approaches. In traffic systems, decisions are made based on current conditions without accurate forecasting. In mining processes, quality is checked after production, which leads to delays in corrective actions. Similarly, in turbofan engines, maintenance is often scheduled or performed after faults occur.

These approaches have several limitations such as lack of real-time prediction, inefficient resource utilization, delayed decision-making, and inability to prevent failures in advance.

Proposed Solution:

The proposed solution uses Machine Learning and Deep Learning techniques to build predictive models for all three problem domains.

For traffic forecasting, time series analysis is used to predict congestion patterns based on historical data. For mining quality prediction, regression models are used to estimate silica impurity levels in advance. For turbofan engines, an LSTM-based deep learning model is used to predict Remaining Useful Life (RUL) using sensor data.

These models help in making proactive decisions, improving efficiency, and reducing operational risks.

Value Addition:

The solution adds significant value by enabling early prediction and better decision-making. It reduces dependency on manual monitoring and improves system reliability. In traffic systems, it helps in planning and congestion control. In mining, it improves product quality and reduces waste. In turbofan engines, it supports predictive maintenance, reducing unexpected failures and maintenance costs.

Overall, the solution enhances efficiency, accuracy, and scalability of industrial operations.

4.1 Code submission (Github link)

Traffic Prediction:

https://github.com/Channaveer06/upskillcampus/blob/main/SmartCity_TrafficPrediction.ipynb

Mining Quality Prediction:

https://github.com/Channaveer06/upskillcampus/blob/main/Mining_QualityPrediction.ipynb

Turbofan RUL Prediction:

https://github.com/Channaveer06/upskillcampus/blob/main/Turbofan_RULPrediction.ipynb

4.2 Report submission (Github link) :

https://github.com/Channaveer06/upskillcampus/blob/main/Channaveer_USC_UCT.pdf

5 Proposed Design/ Model

The proposed system follows a structured machine learning pipeline that includes data collection, preprocessing, feature engineering, model development, and evaluation.

Initially, raw data is collected from different sources such as traffic datasets, mining process data, and turbofan engine sensor data. The data is then cleaned and preprocessed by handling missing values, removing noise, and normalizing the features.

In the next stage, feature engineering is performed to extract meaningful information from the data. For traffic prediction, time-based features such as hour, day, and lag variables are created. For mining, process parameters like air flow and column levels are used. For turbofan prediction, sequential sensor data is prepared for time-series modeling.

After preprocessing, machine learning models are developed. Regression and time-series models are used for traffic and mining predictions, while a deep learning LSTM model is used for turbofan Remaining Useful Life (RUL) prediction.

Finally, the models are evaluated using appropriate performance metrics and used to generate predictions. The overall system ensures accurate, reliable, and scalable solutions for real-world industrial problems.High Level Diagram (if applicable)

5.1 high Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

Raw Data Sources
(Traffic / Mining / Turbofan)

Data Preprocessing
(Cleaning, Normalization)

Feature Engineering
(Time & Sensor Features)

Model Development
(Regression / LSTM)

Model Evaluation
(RMSE, MAE)

Prediction Output
(Traffic / Silica % / RUL)

5.3 Interfaces (if applicable)

Input Data → Processing Module → ML Model → Output Visualization

The system interfaces define how data flows between different components of the machine learning pipeline.

The input interface consists of raw datasets collected from various sources such as traffic sensors, mining process data, and turbofan engine sensor readings. These inputs are provided in structured formats such as CSV files.

The processing interface includes modules for data preprocessing, feature engineering, and model training. This layer handles tasks such as data cleaning, normalization, transformation, and model execution.

The output interface presents the results generated by the models. These outputs include predicted traffic patterns, silica impurity percentage in mining, and Remaining Useful Life (RUL) of turbofan engines. The results can be visualized using graphs, charts, and dashboards.

Additionally, the system can be extended to support real-time data integration and communication with external systems such as IoT platforms and monitoring dashboards, enabling continuous prediction and decision-making.

6 Performance Test

This section evaluates the performance of the proposed machine learning models and demonstrates their effectiveness in solving real-world industrial problems.

The key constraints considered during the design include accuracy, computational efficiency, and data quality. Since the models are trained on real-world datasets, handling missing values, noise, and variability in data was an important challenge.

These constraints were addressed by applying proper data preprocessing techniques such as data cleaning, normalization, and feature engineering. Efficient machine learning and deep learning models were used to ensure good performance without excessive computational cost.

The models were tested using appropriate evaluation metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics helped in measuring the prediction accuracy and reliability of the models.

Although exact real-time deployment constraints such as memory and power consumption were not tested, the models are designed to be scalable and can be optimized further for industrial deployment.

6.1 Test Plan/ Test Cases

The test plan involved dividing the dataset into training and testing sets to evaluate model performance. Different scenarios were considered to ensure that the models perform well on unseen data.

Test cases included validating predictions on different time intervals for traffic data, checking accuracy of silica prediction in mining data, and evaluating RUL prediction for different engine cycles.

6.2 Test Procedure

The testing procedure involved preprocessing the dataset, training the models using historical data, and validating the models on test data.

The models were evaluated using performance metrics such as MAE and RMSE. Predictions were compared with actual values to assess accuracy and reliability.

6.3 Performance Outcome

The performance results indicate that the models are able to capture patterns in the data and generate accurate predictions.

The traffic prediction model successfully forecasted congestion trends, the mining model effectively predicted silica impurity levels, and the turbofan model accurately estimated Remaining Useful Life.

Overall, the results demonstrate that machine learning techniques can significantly improve efficiency, reliability, and decision-making in industrial applications.

7 My learnings

During this internship, I gained valuable hands-on experience in applying machine learning techniques to real-world problems.

I learned how to work with real datasets, perform data preprocessing, and apply feature engineering techniques. I also gained knowledge in time series analysis, regression models, and deep learning approaches such as LSTM.

This internship helped me understand the practical applications of machine learning in industries such as smart cities, mining, and predictive maintenance.

Additionally, I improved my problem-solving skills, analytical thinking, and ability to work on end-to-end machine learning projects.

Overall, this internship has strengthened my foundation in Data Science and prepared me for future career opportunities in this field.

8 Future work scope

The current system can be further improved by enhancing the accuracy and efficiency of the machine learning models used in all three projects.

For the traffic forecasting system, future work can include incorporating real-time traffic data and advanced time series models to improve prediction accuracy and adaptability.

In the mining quality prediction system, more advanced algorithms and larger datasets can be used to improve the prediction of silica impurity and optimize the overall process.

For the turbofan engine prediction system, more sophisticated deep learning models and additional sensor data can be utilized to achieve more accurate Remaining Useful Life (RUL) predictions.

The models can also be deployed as real-time applications using web-based platforms or cloud services, enabling continuous monitoring and prediction.

Integration with IoT systems can further enhance the system by providing live data inputs and improving decision-making in real-world industrial environments.

Overall, future improvements will focus on increasing accuracy, scalability, and real-time implementation of the proposed solutions.